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Reconstruction of bisphenol A intake using a simple pharmacokinetic model

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Abstract

Bisphenol A (BPA) is used in the manufacture of a range of consumer products, and human biomonitoring studies suggest that exposure to BPA is nearly ubiquitous. We constructed and calibrated a simple pharmacokinetic model to predict urinary concentrations of BPA based on a known initial dose. This descriptive (rather than physiologically based) model has three compartments: “stomach/liver,” “blood,” and “bladder.” We calibrated and validated the model parameters using blood and urine measurements from nine volunteers who consumed 5 mg of d₁₆-BPA. We then applied the model to a second group of eight persons, who supplied full volumes of urine over 7 consecutive days and a diary identifying times and types of food and beverage consumed, to “reconstruct” the time and mass of BPA intakes. These reconstructed daily intakes ranged on average from 60 to 100 ng/kg-day, within the range of, but slightly higher than, those surmised from other studies. About two-thirds of intakes occurred within an hour of reported food or drink consumption, supporting the hypothesis that diet is the main pathway of exposure to BPA. However, one-third of all reconstructed intakes took place outside this time window, suggesting that other sources of BPA exposure may also be relevant.

Keywords

biomonitoring; BPA; exposure; metabolism; pharmacokinetic

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

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INTRODUCTION

Bisphenol A (BPA) is a high-volume industrial chemical, with an estimated US production volume of 2.4 billion tons in 2007.¹ BPA is used in the manufacture of epoxy resins and polycarbonate plastics, which are used in a range of consumer products (e.g., cans and other food containers, thermal paper, medical devices).² Consequently, human biomonitoring studies have demonstrated that exposure to BPA is nearly ubiquitous.³ Controlled dosing studies suggest that BPA is virtually completely eliminated in urine, with a relatively short half-life of approximately 4–5 h.^{4–7}

Exposure to BPA has generally been estimated in one of two ways. The first method, termed the “forward approach,” combines BPA concentrations in exposure media (e.g., air, drinking water, food) with exposure media contact rates. Researchers using this approach have concluded that diet is the major route of general population exposure to BPA.^{7–11} Other sources of exposure may include some dental sealants,¹² paper products,^{13,14} dust,^{11,15,16} medical interventions,^{17–19} and indoor air.^{9,11} Using the forward approach, estimated intakes range from 1.4 $\mu\text{g}/\text{kg}\cdot\text{day}$ for adults consuming canned foods (a high exposure scenario)⁸ to 0.03–0.06 $\mu\text{g}/\text{kg}\cdot\text{day}$.^{9,10} With the second method to estimate BPA exposure, termed the “backward approach” one estimates the intake “dose” or external exposure, which would have produced the observed BPA concentrations in tissues or body fluids (generally urine). This approach provides an estimated BPA intake which is integrated over all (known and unknown) exposure sources and routes. Studies using the backward approach applied to BPA urine concentrations estimate average intakes ranging from ~0.02 to 0.03 $\mu\text{g}/\text{kg}\cdot\text{day}$ among adults in the United States and Japan^{7,20–22} to 0.04 $\mu\text{g}/\text{kg}\cdot\text{day}$ among students (aged 20–29 years) in Germany,²³ to 0.07 $\mu\text{g}/\text{kg}\cdot\text{day}$ among US girls aged 6–8 years.^{7,24}

An additional approach to intake estimation relies on pharmacokinetic (PK) modeling. Use of a PK model is a “forward” approach, but in contrast to the approach described above, also incorporates biomonitoring data. With the proper data, a PK model can be developed and calibrated to predict tissue concentrations (e.g., urine) given a specified intake regime. Conversely, given a known excretion, the PK model may be used to estimate the intake profile (time and mass) necessary to produce the observed excretion. The biomonitoring data requirements for this approach can be daunting, including not only accurate measurements of the target contaminant in the appropriate tissue or body fluid, but also characteristics such as the time and volume of urine void events or serum collection. Lorber and Calafat applied this approach to study exposure patterns for di(2-ethylhexyl) phthalate (DEHP) relying on biomonitoring data for eight individuals with no known occupational exposure to DEHP, over the course of a normal week.²⁵ In this paper, we use a similar approach to study BPA exposure. First, we develop, calibrate, and test a model, which predicts BPA concentrations in urine using data from a controlled dosing human study. Then, we apply the model to the same group of eight persons studied by Lorber and Calafat²⁵ to describe patterns of exposure to BPA, including estimates of total daily intakes, number of intake events per day, and the relationship between intakes and food consumption.

MATERIALS AND METHODS

We used controlled dosing data from four adult individuals to construct a three-compartment PK model and then validated the model using additional data (six individuals) from the same study.⁵ Finally, we applied the model to a group of eight persons who collected all urine voids for 7 consecutive days, with diary data recording timing and details of food and beverage consumption as reported by Lorber and Calafat.²⁵ Each step is described in further detail below.

Stage 1: Model Fitting

The controlled dosing data come from nine volunteers who were administered 5 mg of d₁₆-BPA (Table 1) to rule out background exposure to the ubiquitous BPA.⁵ The dose (>50 µg/kg-day, depending on body weight) was substantially higher than estimated background BPA exposure (generally <0.1 µg/kg-day). The study was conducted in two parts. First, four male volunteers (ranging in age from 28 to 54 years) provided blood samples at nine time points from 0 to 381 min. We used these d₁₆-BPA blood concentrations to fit our PK model and estimate model parameters. Details on this model calibration step are provided below. In the second part of the study, six volunteers (three male and three female, ranging in age from 24 to 32 years of age) provided blood (collected at six time points from 0 to 24 h) and urine (eight time points from 0 to 40 h). We used the urine and blood d₁₆-BPA concentrations in this second part to validate our model. One volunteer (subject G) participated in both the first and second parts of the study. The Regional Ethical Committee of the University of Würzburg approved the study and volunteers provided written informed consent as described.⁵

We show our three compartment model in Figure 1. Ingested d₁₆-BPA goes first to the intestinal tract, and then diffuses into the bloodstream. It then passes to the bladder before being excreted in the urine. This structure is not intended to be physiologically rigorous - rather it is an empirical approach designed to fit the experimental data. For example, the fact that blood concentrations of d₁₆-BPA at the second collection (~2 h) were higher compared with those measured at ~1 h suggested that ingested d₁₆-BPA did not immediately deposit into a blood reservoir, and hence, we designed a holding reservoir, which we conveniently termed the “stomach/liver” in Figure 1. Similarly, passage from a “blood” reservoir to a “bladder” reservoir is not necessarily physiological, but we created these reservoirs to fit the available experimental blood and urine concentrations of d₁₆-BPA. The five parameters that describe this first-order mass balance model are: apparent blood volume of distribution (B , l/kg), two fractional constants representing the fraction of d₁₆-BPA available to transfer from stomach/liver to blood ($F1$) and from blood to bladder ($F2$), and two transfer rate constants representing transfer of d₁₆-BPA from stomach/liver into blood ($K1$, 1/min) and from blood to bladder ($K2$, 1/min). We implemented this model in an Excel spreadsheet where each row represents a 5 min time step in the model. We show the numerical equations coded into this spreadsheet in Table 2.

Volkel *et al.* measured d₁₆-BPA-glucuronide (d₁₆-BPA-G) in the blood and urine of the volunteers.⁵ The liquid chromatography/mass spectrometry (LC/MS) method developed as part of the study and used to measure d₁₆-BPA-G also measured free d₁₆-BPA in blood and

urine. They did not detect any free d₁₆-BPA in either plasma or urine after oral administration of d₁₆-BPA, even in samples with rather high concentration of d₁₆-BPA-G, suggesting that d₁₆-BPA was completely metabolized to d₁₆-BPA-G after ingestion. To test this hypothesis, they conducted a side study with a subset of volunteers. Specifically, they obtained blood samples at short intervals from the first four volunteers; identified as subjects G, M, N, and O. In addition to measuring d₁₆-BPA-G by LC/MS, the researchers measured total d₁₆-BPA (d₁₆-BPA-G plus free d₁₆-BPA) by cleaving the glucuronide by deconjugation with glucuronidase using gas chromatography/mass spectrometry. They found that the total d₁₆-BPA concentration quantified by gas chromatography/mass spectrometry matched well with the d₁₆-BPA-G concentration quantified by HPLC/MS/MS, when both concentrations were expressed on a molar basis (Pearson correlation coefficient = 0.97). This supported the hypothesis that d₁₆-BPA completely metabolized to d₁₆-BPA-G after ingestion.

This also allowed two sets of measurements to calibrate the model - one set with d₁₆-BPA-G and one with total d₁₆-BPA. We sought to determine a set of calibration parameters that would provide an adequate fit for both sets of blood data; one was not favored over the other. In the calibration, we established the 5-mg dose of d₁₆-BPA in terms of molar mass as follows: $5 \text{ mg} \times (1 \text{ g}/1000 \text{ mg}) \times (1 \text{ mole}/244 \text{ g}) \times (1\text{E}^{12} \text{ picomole}/\text{mole}) = 2 \times 10^7 \text{ picomoles}$, where 244 g is the molecular weight of d₁₆-BPA; an equivalent mass of d₁₆-BPA-G was assumed. We simulated the fate of the mass in terms of picomoles, and then converted back to mass units using the molecular weights of the two species: 244 g/mole for d₁₆-BPA²⁶ and 418 g/mole for d₁₆-BPA-G.⁵ We obtained an initial estimate for the parameters using the built-in Excel Solver to minimize the summed absolute difference (over all blood collections after t₀ and all four subjects) between predicted and observed blood concentrations of d₁₆-BPA-G and total d₁₆-BPA. We then manually adjusted the parameters to arrive at a visual “best-fit” for the data. A second adjustment step was necessary because the Excel Solver best-fit solution identified a peak concentration that occurred earlier than the data suggested. In using Excel Solver and then adjusting manually, we constrained the parameters as follows: apparent blood volume of distribution, *B*, could range from 0.1 to 1.0 l/kg, and *K*₁ and *K*₂ could take any value ≥ 0.011/min. The calibrated parameter for apparent blood volume of distribution is not physiologically equivalent to a volume of blood circulating at any given time; rather it is a “mixing volume.” Previous modeling studies of DEHP^{25,27} and perchlorate,²⁸ for example, reported blood volumes of distribution to be about 0.1 and 0.34 l/kg, respectively. The restriction on the rate parameters essentially insures that they be assigned non-zero values. Ingested BPA appears to be excreted fully in urine.²⁹ Indeed, Volkel *et al.* reported recovering 118% (±21%) of the administered dose of d₁₆-BPA in urine, suggesting an exceedance of mass eliminated compared with mass ingested.⁵ For this reason, we set *F*₁ and *F*₂ at a fixed value of one, to represent the expected, and indeed, observed elimination kinetics (i.e., full excretion of the ingested dose of d₁₆-BPA in urine).

From this point forward, model predictions for this experimental cohort will be referred to as, “d₁₆-BPA.” As noted earlier, all other measurements, including those for the model validation described next, were for d₁₆-BPA-G. Predictions and measurements will be on an equivalent basis as both are expressed on a molar basis.

Stage 2: Model Validation

After fitting the model using the total d_{16} -BPA and d_{16} -BPA-G blood concentrations from four volunteers, we applied the model to the blood and urine d_{16} -BPA-G concentrations provided by six volunteers.⁵ We plotted and compared the observed concentrations of d_{16} -BPA-G and predicted concentrations of d_{16} -BPA.

Stage 3: Model Application

We applied the fully parameterized model to a data set previously described, in which eight adult volunteers (four men and four women, aged 25–59 years) provided a complete set of urine spot samples ($n = 427$) over a consecutive 7-day period in October–November of 2005.^{30–32} Each participant collected every urine void in a commercial, non-vinyl and non-polycarbonate plastic specimen collection cup. After recording the volume of each void, the urine was decanted in a prelabeled, sterile urine cup and stored in an ice cooler. The urine samples were retrieved from participants daily (or after the weekend), aliquoted into polypropylene cryovials or glass jars, and stored at -70°C until analysis. During the study week, each volunteer also recorded all food and beverage consumed (and time of consumption), medications consumed (if any), and limited data on other activities such as driving. The Institutional Review Board of the Centers for Disease Control and Prevention approved the study and all participants provided written informed consent.

The total (free plus conjugated) BPA concentrations in urine were determined using on-line solid phase extraction coupled to isotope dilution-high-performance liquid chromatography-tandem mass spectrometry at Centers for Disease Control and Prevention. The limit of detection (LOD) was 0.4 ng/ml; values below the LOD were replaced with a value (0.28 ng/ml) equal to the LOD divided by the square root of 2. To ensure data accuracy and precision, each batch of study samples included two high concentration (~ 10 ng/ml) and two low-concentration (~ 2.5 ng/ml) quality control samples (QCH and QCL, respectively), analytical standards, and reagent blanks. The method accuracy, expressed as a spiked recovery percentage, ranged from 98% to 113% at four different spiking levels.³³ The method precision, determined by calculating the coefficient of variation of 22 repeated measurements of the QCL and QCH materials over 6 months, was 7.4% (QCL) and 5.3% (QCH).

We set up the model for each volunteer, and assigned BPA intakes to time-steps based on concordance between the observed and predicted urinary concentrations (molar basis) of BPA. After assigning BPA intake times and quantities, we then looked at these reconstructed intakes in relation to reported food and beverage intake as reported in the volunteer diaries.

RESULTS

Stage 1: Model Fitting

The model fitting process led to the parameter estimates shown in Table 3. The fitted value for blood volume was 0.15 l/kg, whereas the fitted values for the elimination constants were 0.9 1/min. Overall, the predicted values showed good agreement with the observed blood concentrations from the four volunteers (Figure 2) of the original study,⁵ with a Pearson

correlation coefficient of ~0.8 between predicted and observed blood concentrations (0.80 for d₁₆-BPA and 0.78 for d₁₆-BPA-G). As discussed in the Materials and methods section, part of the calibration involved a “visual best-fit” to ensure that both the time and the magnitude of peak concentrations were adequately modeled. As seen in Figure 2, peak concentrations were ~750 pmol/ml for three of four individuals, reached at 1–2 h (the fourth individual experienced higher peak values >1000 pmol/ml). The model predictions, shown as the smooth curves in Figure 2, reached a peak at about 750 pmol/ml at about 1 h. Note that the observed data show heterogeneity in individual response to the elevated dose of d₁₆-BPA, but the modeled response is identical for each individual, because the model does not rely on any individual-specific parameters (such as age or weight). Certainly individual rate constants could be developed, but our goal was to determine a single set of parameters that could be used for general population application. Based on our visual best-fit (and the reasonable Pearson correlation coefficient), we believe our model adequately portrays the “average” rise and decline of BPA in serum.

We then applied the model to the six subjects with blood and urine concentration data (examples for two subjects shown in Figure 3, data for all subjects shown in Supplementary Figures 1 and 2). The model tended to slightly under- or over-predict urine concentration at the first time point, but matched well with observed concentrations at subsequent time points. The Pearson correlation coefficients for predicted and observed concentrations were 0.87 (blood) and 0.97 (urine). Note that because the first experimental time point was 5 h, the peak concentrations could not be observed for this second set of volunteers and thus were modeled.

Stage 3: Model Application

In the last stage, we applied the model to each of eight volunteers for whom we had the BPA concentrations for all their urine voids over 7 consecutive days, in conjunction with a diary detailing their food and beverage consumption. We fit the BPA exposure events (amount and timing) by trial-and-error to produce predicted BPA mass excretions close to the observed mass urinary excretions. An example for one subject is shown in Figure 4, displaying in essence the near-perfect match between the predicted mass excreted compared with the measured excreted mass that was, of course, obtained by design. Similar results are found with all other subjects (data not shown). Overall, the Pearson correlation coefficients between expected and observed mass urinary excretion ranged from 0.92 to 0.99 across the eight subjects. This resulted in estimated average intakes on the order of 60–100 ng/kg-day (range: 15.9–347.0 ng/kg-day; Table 4). As a check to our approach, we also calculated the BPA excretion for each individual over the course of the week based on measured concentrations and urine volumes. Assuming that intakes equal excretion, mass excretions would equal mass intakes. We found a very close match between each person’s reconstructed intakes and their excretions (Table 5). The absolute difference between the volume-based excretions and reconstructed intakes ranged from 0.2% to 11.1%.

All eight individuals excreted BPA on each of the 7 consecutive days, suggesting that exposure occurred every day. On average, there were four to seven exposure events reconstructed for a given individual any given day. The reconstructed events were relatively

spaced out over each day, although the amount of BPA ingested did vary somewhat over the course of the day. In general, the average intake was highest during the 1800 hours, and was higher during the day compared with hours when most subjects would be asleep. Figure 5 shows the average hourly intakes of BPA calculated over the entire cohort (bar at “1” indicates exposures occurring between midnight and 0100 hours and so on). There was one particularly high reconstructed intake (15,550 ng or ~259 ng/kg, based on a body weight of 60 kg), which led to a high average for the 0100 hours. This single event occurred 1.5 h after the individual reported consuming green tea with mint, and 2 h before a 2.5-mile walk. The other reported consumption events occurring that day were breakfast foods (cereal and milk, coffee) consumed 7 h before the reconstructed dose, a large diet Coke consumed 2.5 h afterward, and dinner foods (pasta dish containing chicken, vegetables, and cheese) consumed 6.5 h afterward. This individual also reported taking several vitamin supplements each morning, and a prescription medication each evening, during the study period (exact times were not noted).

Having used this approach to validate the PK model, we then looked at the reconstructed intakes in relation to each participant's recorded diary data. In some instances, we found good correlations between the timing of reconstructed intakes and reported food and drink consumption. For example, for one individual on one day, all the reconstructed BPA intakes occurred at or close to the time the participant consumed food or drinks (Table 6). In other instances, we did not find a clear link between reconstructed intakes and reported food and beverage consumption. For example, Table 7 shows similar results for another individual where the reconstructed intakes at 0215 hours and 1015 hours did not correspond to any reported food consumption. This could be attributed to individual differences in metabolism, to unreported consumption of food or beverages, or to other non-dietary activities that may have resulted in exposure to BPA.

DISCUSSION

We sought to construct, validate, and apply a simple three-compartment PK model to predict urinary concentrations of BPA. We used data from nine human volunteers who ingested d₁₆-BPA⁵ to construct and validate the model, and then applied it to a group of eight volunteers, from a different study, who collected all urine voids over a 1-week period.^{30,31} We also reconstructed the BPA intakes for these eight individuals, and related these to recorded food and beverage consumption. The estimated median (mean) BPA intake for these eight persons was 0.06 (0.08) $\mu\text{g/kg-day}$, which is similar to or somewhat higher (albeit of the same order of magnitude) than estimates from previous studies in Japan, the United States, and Germany (Table 8). The higher intake estimated in this study could be related to several factors including differences in study populations (e.g., age range and geographic location) and exposure patterns; or the method for estimating intake (i.e., the construction and calibration of the PK model). For example, Lakind and Naiman,²¹ used spot sample concentration data from the National Health and Nutrition Examination Survey combined with generic values for daily urine output (1.6 l/day for men, 1.2 l/day for women) and reported a median intake for the US population of 0.03 $\mu\text{g/kg-day}$. However, these urine outputs are lower than the average daily urine outputs across the eight subjects collecting all urine voids for 7 days, which ranged from 1.41 to 2.13 (men) and 1.46 to 3.58 (women) l/

day. Thus, had Lakind and Naiman²¹ used urine output values more similar to those found in our study, they may have arrived at more similar intake estimates as well.

Diet has been cited as the primary source of non-occupational BPA exposure.^{7,34,35} Our analysis supports that contention, in that for the 325 fitted exposure events, the majority of them, 63.7%, occurred within 1 h of reported food or drink consumption. However, 36.3% ($n = 118$) of the exposure events occurred outside of this time window, that is, not concurrent with food or drink consumption (as noted in the example of subject 3). This is similar to the results reported by Lorber and Calafat²⁵ in their study of reconstructed DEHP intake within this cohort, where they found that one-third of reconstructed DEHP intakes were outside a 2-h window. They also found that ~20% of exposures occurred during the evening hours when individuals might be asleep. One possibility is that there were unreported food or beverage intake events; which could influence our conclusions. For example, if there were some unreported midnight snacking, this may have influenced our conclusions regarding nighttime exposures. Lorber and Calafat²⁵ also suggested that nighttime exposures to DEHP might have been related to inhalation of airborne residues, use of medical items such as continuous positive air pressure machines for sleep apnea or a mouth guard, emission of DEHP from vinyl materials (e.g., flooring) in the participant's home, or possibly use of personal care products.²⁵ These authors also hypothesized that the night exposure events may be related to the release of DEHP metabolites that resided in storage depots in the body. All of the above scenarios might also explain the nighttime BPA exposures. Importantly, BPA exposure from routes other than ingestion (e.g., dermal, inhalation), or exposure from medications, may not be metabolized in the same way that BPA ingested from food and beverage (e.g.,³⁶), and may not be adequately described by our model.

Although this study developed a simple PK model for BPA, previous studies have described more complex physiologically based pharmacokinetic models (e.g.,^{37–39}). An advantage of these physiologically based pharmacokinetic models is that they are generally much more detailed. However, their development and use is even more data intensive than the model we describe, and usually rely on extrapolation from experimental animal studies. In contrast, our model is easily implemented on an Excel spreadsheet and does not require knowledge (or extrapolation) of many physiologic parameters. Thus, it may be an attractive option when the goal is to gain understanding of general exposure levels and patterns, without the need for very detailed data. However, we recognize that the model was developed using a very small data set, and thus may not be applicable to all human populations.

CONCLUSIONS

We developed a three-compartment PK model using blood concentration data from four volunteers who ingested d₁₆-BPA in a controlled dosing study and validated the model using blood and urine concentration data from six additional volunteers in the same study. We then applied the model to a “real world” data set to reconstruct BPA exposures. The daily intakes derived in this dose reconstruction exercise are consistent with extrapolated intakes based on urinary concentration of BPA, and with intake estimates reported in other studies. Analysis of the specific reconstructed intake events supports an association between food consumption and BPA exposure, although some modeled intakes occurred at nighttime when

no food consumption was reported. Although data intensive, both in terms of data to develop and also to apply the model, this exercise shows the potential for use of simple PK models to study patterns of chemical exposure in the general population.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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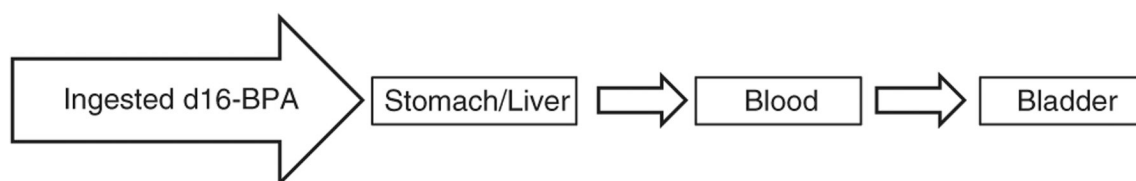


Figure 1.
Three compartment pharmacokinetic model for d₁₆-bisphenol A (BPA).

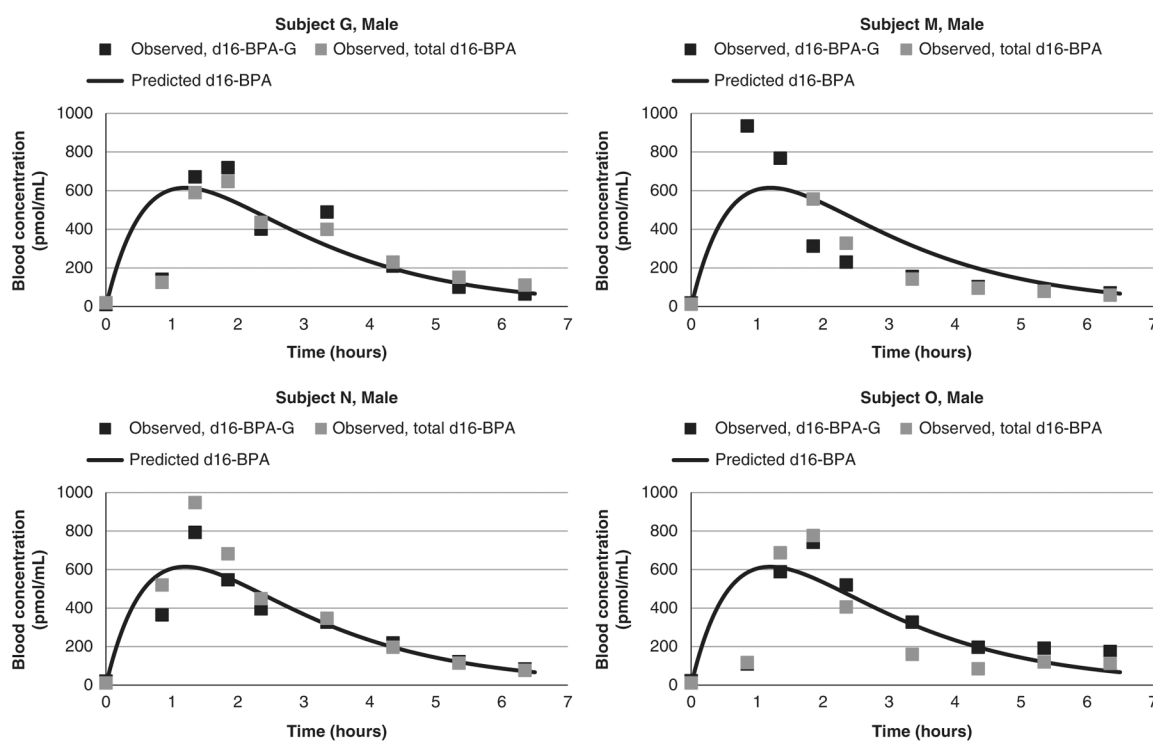


Figure 2.

Observed and predicted concentrations of d₁₆-bisphenol A (BPA) in blood of four volunteers. The observed values were calculated from the data reported by Volkel *et al.*⁵

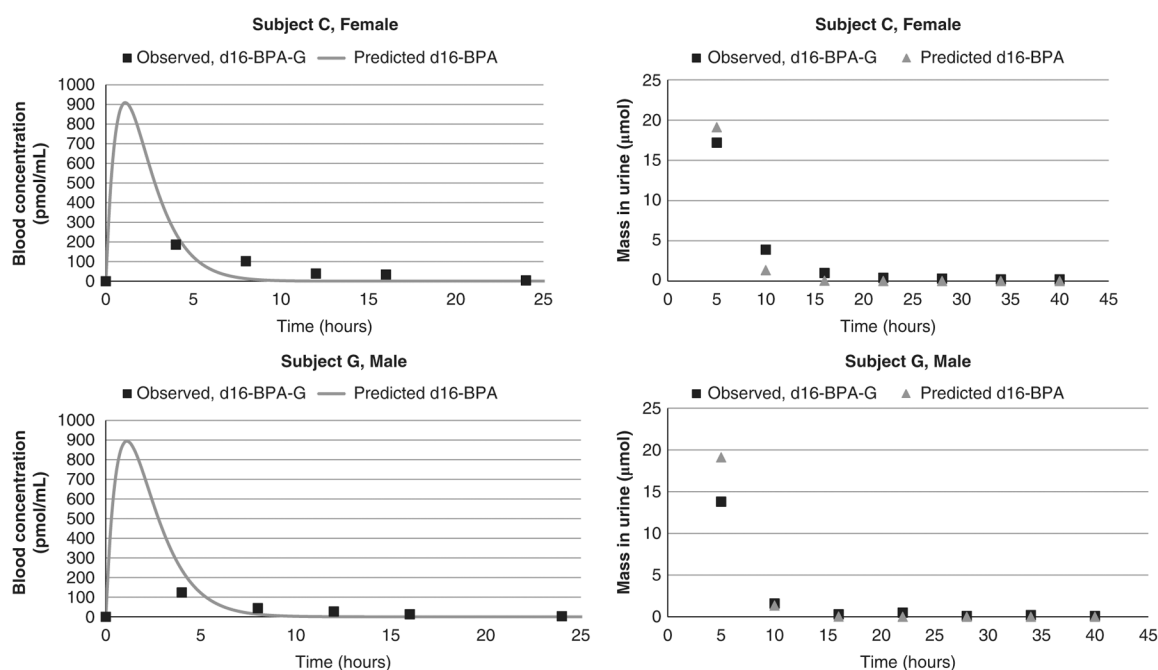


Figure 3. Observed and predicted concentrations of d₁₆-bisphenol A (BPA) in (a) blood and (b) urine of two volunteers. The observed values were calculated from the data reported by Volkel *et al.*⁵

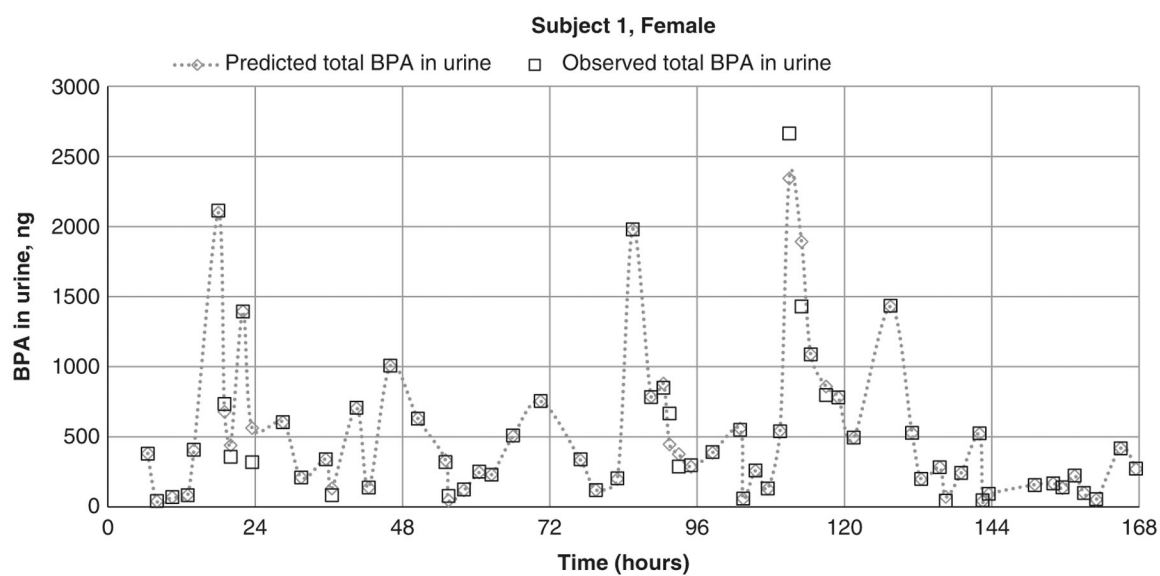


Figure 4.

Observed and predicted mass excretion of bisphenol A (BPA) in urine of two volunteers.

The observed values were calculated from the data reported by Preau *et al.*³⁰ and Ye *et al.*³¹

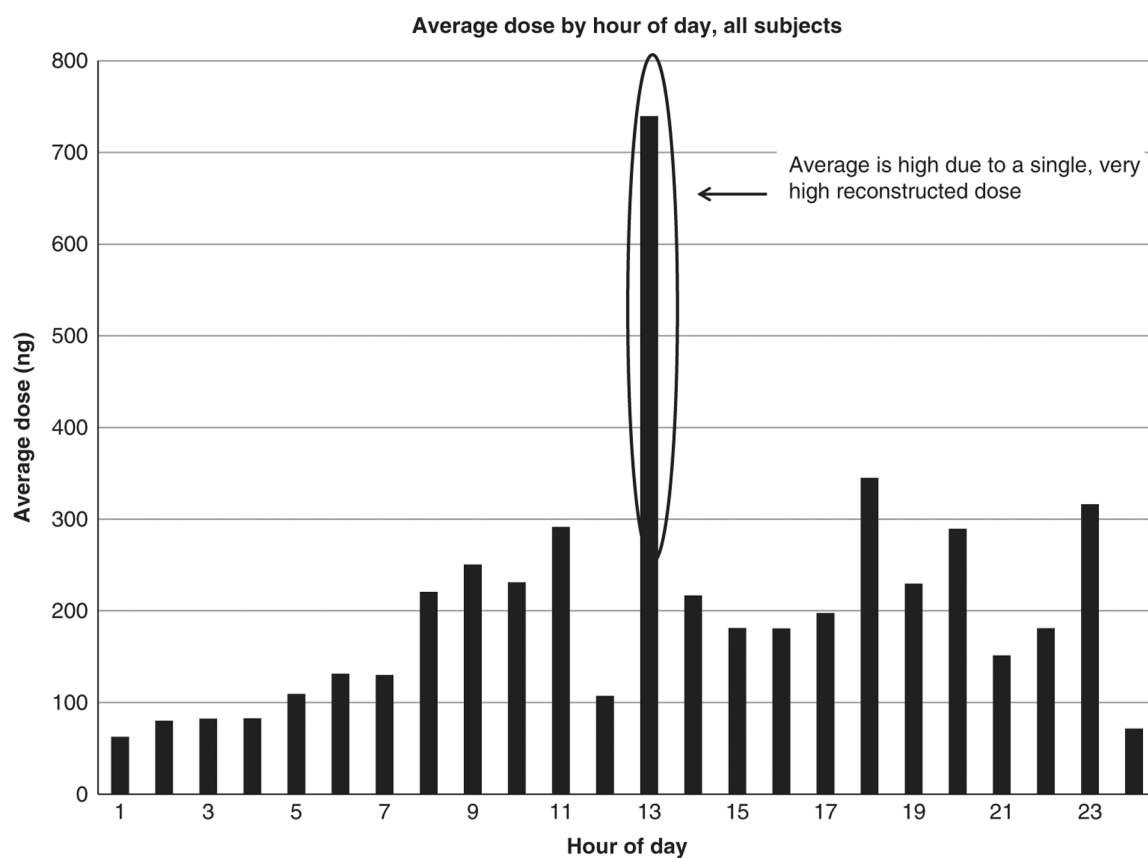


Figure 5.

Average bisphenol A (BPA) intake for each hour of the day for the eight volunteers studied by Preau *et al.*³⁰ and Ye *et al.*³¹

Table 1.Demographic characteristics of participants in the controlled dosing experiment.⁵

Subject	Sex	Age (years)	Height (cm)	Weight (kg)
<i>Detailed blood data: Blood measurements of d₁₆-BPA and d₁₆-BPA-G at 0, 51, 81, 111, 141, 201, 261, 321, and 381 min (concentration, pmol/ml)</i>				
G	Male	28	176	63
M	Male	47	188	83
N	Male	54	182	92
O	Male	39	179	74
<i>Blood and urine data: Blood measurements of d₁₆-BPA-G at 0, 4, 8, 12, 16 and 24 h (concentration, pmol/ml), and urine measurements of d₁₆-BPA-G at 5, 10, 16, 22, 28, 34 and 40 h (mass, µmol)</i>				
A	Female	29	165	57
B	Female	24	175	63
C	Female	31	162	62
E	Male	29	184	82
F	Male	30	175	73
G	Male	28	176	63

Abbreviation: BPA, bisphenol A.

Table 2.

First-order equations describing the three-compartment pharmacokinetic model.

Mass in compartment at time t (pmol)				
	Current mass	Previous mass	Incoming	Outgoing
Stomach	$M_{S,t} =$	$M_{S,t-1}$	Ingested d_{IG} -BPA	$- M_{S,t-1} * (1 - EXP(-K1 * \Delta t))$
Blood	$M_{B,t} =$	$M_{B,t-1}$	$+ (F1 * M_{S,t-1}) * (1 - EXP(-K1 * \Delta t))$	$- M_{B,t-1} * (1 - EXP(-K2 * \Delta t))$
Bladder	$M_{U,t} =$	$M_{U,t-1}$	$+ (F2 * M_{B,t-1}) * (1 - EXP(-K2 * \Delta t))$	

Abbreviation: BPA, bisphenol A.

Table 3.

Model parameters.

Parameter	Name	Fitted value
Apparent blood volume of distribution (l/kg)	B	0.15
<i>Fractional constants</i>		
Fraction d ₁₆ -BPA from stomach to blood	F1	1.0 (Fixed)
Fraction d ₁₆ -BPA from blood to bladder	F2	1.0 (Fixed)
<i>Elimination constants (1/min)</i>		
Elimination of d ₁₆ -BPA from stomach to blood	K1	0.9
Elimination of d ₁₆ -BPA from blood to bladder	K2	0.9

Abbreviation: BPA, bisphenol A.

Table 4.

Estimated BPA daily exposure for eight volunteers.^{30,31}

Subject	Daily exposure (ng/kg-day) ^a		
	Mean (SD)	Min	Max
1: Female, 31 years	76.3 (43.6)	24.8	151.2
2: Female, 25 years	67.1 (21.6)	46.2	111.3
3: Female, 59 years	101.1 (111.2)	29.8	347.0
4: Female, 27 years	85.6 (50.6)	47.8	192.7
5: Male, 34 years	67.3 (29.0)	33.9	120.0
6: Male, 26 years	96.2 (53.1)	36.3	170.4
7: Male, 32 years	70.7 (56.4)	34.1	194.7
8: Male, 32 years	58.5 (52.6)	15.9	156.1

Abbreviation: BPA, bisphenol A.

^aAssumes 60 kg body weight for women, and 70 kg body weight for men.

Table 5.

Estimated BPA daily exposure for eight volunteers,^{30,31} comparing two methods.

Subject	Total BPA exposure over study period (µg)		
	A	B	C
	$\Sigma V \text{volume} \times \text{concentration}$) "Observed"	PK model ("Predicted")	% Difference between (A) and (B) ^a
1: Female, 31 years	31.6	32.0	-1.5
2: Female, 25 years	25.4	28.2	-11.1
3: Female, 59 years	41.4	42.5	-2.6
4: Female, 27 years	34.5	36.0	-4.3
5: Male, 34 years	31.6	32.3	-4.5
6: Male, 26 years	47.1	47.2	-0.2
7: Male, 32 years	35.7	34.7	2.9
8: Male, 32 years	29.2	28.7	1.9

Abbreviation: BPA, bisphenol A.

^a% difference calculated as $100 \times [(\text{observed} - \text{predicted}) / \text{observed}]$.

Table 6.

Relationship between reconstructed BPA exposures and recorded food and beverage intake for Subject 4, Day 2 (weekday).

Time	Food and drinks	Predicted BPA dose (ng)
0100	Ice cream sandwich	
0130		450
0200	8 oz Tap water	
0300	8 oz Tap water, 1 Ibuprofen (600 mg)	580
0830	8 oz Water	840
1000	1 Reese's Cup, 12 oz water	
1030	1 Small banana, 12 roasted/unsalted almonds	
1130	Hot tea, water	
1145		1250
1200	Hot and sour soup, garlic chicken, sweet and sour chicken, egg roll w/ spicy mustard	
1330	1 Mini Reese's Cup, 10 green grapes	90
1400	12 oz water, 4 strawberries, 12 roasted/unsalted almonds	
1530	Piece of chocolate, 12 oz coffee w/ powdered cream and sweet n' low	
1730	1 Reese's Cup, 10 green grapes	390
1800	6 oz Water	
1845		1400
1900	Chicken w/ cream of mushroom soup and rice	
1930	8 oz Water	
2300	10 Green grapes, 2 tbsp lite tiramisu ice cream	940

Abbreviations: BPA, bisphenol A; w/, with.

Table 7.

Relationship between reconstructed BPA exposures and recorded food and beverage intake for Subject 8, Day 6 (weekend).

Time	Food and drinks	Predicted BPA dose (ng)
0030	Bowl of ramen noodles (oriental flavor)	240
0215		250
1015		640
1030	Cup of coffee w/ sugar and cream	
1230	Caramel flavored rice cake w/ peanut butter, orange juice	
1300	50 mg Zinc, 800 mg folic acid	
1530	20 Bite-size tortilla chips	
1730	1 Serving of scalloped potatoes	380
1900	1 Small piece of sausage	
2130	10 Buffalo wings, fries, celery, ranch dressing, water	

Abbreviations: BPA, bisphenol A; w/, with.

Table 8.

Estimated BPA intake per day from various studies.

Country	Estimated intake (µg/kg-day)	Reference
Germany, infants 1–5 months	0.02–0.07	40
Germany, children 3–5 years	0.12	41
USA, children 6–11 years	0.07	7
USA, girls 6–8 years	0.07	24
Japan	~0.02 to 0.03	22
USA, general population	0.03	21
USA, general population	0.05	42
Germany, students 20–29 years	0.04	23
USA, adults 25–59 years	0.08 (mean), 0.06 (median)	This study

Abbreviation: BPA, bisphenol A.